CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. Examination (CE)

May 2014

CE214 MICROPROCESSOR & INTERFACES

Date: 10.05.2014, Saturday **Time: 10:00 am To 01:00 pm** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections. Section I and II must be attempted in separate answer sheets.
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of scientific calculator is allowed.

SECTION-I

Q-1 Answer the following questions. **[11]**

[A] How many address lines are necessary to address 2MB of memory? [1]

[B] What determines that Microprocessor is an 8, 16 or 32 bit? [1]

[C] List register pairs in 8085.What is the specialty of HL pair over the other register pairs? [1]

[D] What is use of RESET OUT and ALE pins of 8085? [1]

[E] Write down the operation of chip given in Figure 1(Page No 3). [1]

[F] What is minimum pulse width required for INTR signal? [1]
(Note: CALL instruction takes 18 T states and Clock speed is 3 MHz)

[G] List the machine cycles needed to execute STA 2050h. [1]

[H] Which one of the following is non-maskable interrupt? [1]
a) INTR b) INT c) TRAP d) RST

[I] ________ is a level-sensitive interrupt. [1]
a) RST 7.5 b) INT c) TRAP d) RST 5.5

[J] In the I/O mapped I/O, can an input and an output port have the same port address? Why? [1]

[K] When does microprocessor check for interrupt? [1]
a) after completion of program execution
b) at start of program execution
c) at the end of execution of current instruction
d) at the end of execution of current T-state of current instruction

Q.2 Answer the following questions. **[12]**

[A] What are the internal components of 8155 (Multipurpose Programmable Device)? Why is it called MPD? Draw diagram only for interfacing 8155 with 8085. [5]

[B] Draw and explain the timing diagram for LXI B, 4000H. (Opcode for LXI B instruction is 01H). [7]

OR

[B] 8085 has five types of interrupt. Write down characteristic of each type of interrupt. Write down the eight steps of serving 8085 interrupt. [7]

Q-3 Answer the following questions.(Attempt any Two) **[12]**

[A] How many flags used by 8085? How does flag affect execution of program?

[B] How does 8085 perform internal data operations? How many registers it has to perform operation? Write functionality of each register.

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[C] With a neat diagram explain the interfacing circuit using 74138 (3-8 Decoder) needed to connect the following with 8085 microprocessor address lines.

a. 2K × 8 RAM chip in the address space C000H-C7FFH,
b. 4K × 8 RAM chip in the address space D000H-DFFFH,
c. 2K × 8 EPROM chip in the address space 0000H-07FFH,
d. 8K × 8 EPROM chip in the address space 2000H-3FFFH,

In this case, do you have multiple memory address ranges for any of the memory chips? If so, identify the different equivalent address ranges.

SECTION-II

Q-4 Answer the following questions. **[11]**

[A] What is the difference between INR & INX instructions? [1]

[B] If the 8085 adds 87H and 79H, specify the contents of the accumulator and the status of the S, Z, AC, P and CY flags. [1]

[C] Which one of the following is true for HLT instruction? [1]

a) This is two byte instruction.
b) The processor restarts execution.
c) The address bus and data bus are placed in high impedance state.
d) None of the above

[D] If the clock frequency is 5 MHz, how much time is required to execute an instruction of 18 T-states? [1]

[E] Which one of the following is wrong instruction? [1]

a) LXI B
b) LDAX C
c) ADD B
d) ORA A

[F] __________ flag will not get affected in INR and DCR Instruction. [1]

[G] Write down meaning of instruction only: RAL, RRC [1]

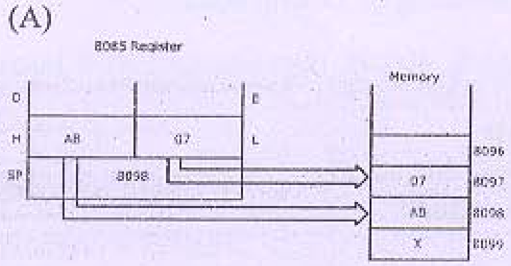
[H] When a subroutine is called, the address of the instruction following the CALL instruction is stored in/on the [1]

a) Stack Pointer
b) Accumulator
c) Program Counter
d) Stack

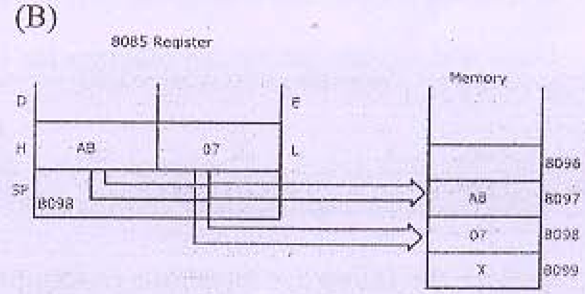
[I] Which one of the following is true for following instruction? [1]

LXI H, AB07H
PUSH H

(A)



(B)



[J] Calculate the number of T-states required to execute following program [1]

	MVI B, 38H	7 T
LOOP:	DCR B	4 T
	JNZ LOOP	10/7 T

[K] What is the role of W & Z register in CALL & RET instruction execution? [1]

Q-5 Answer the following questions. **[12]**

[A] Write down Opcode, Operand and Description of Arithmetic operations supported by 8085. [3]

[B] By writing suitable program, describe the working of the instruction CALL and RET. [4]

[C] Write an assembly language program along with flow chart to exchange 10 bytes of data stored from location 4500h with 10 bytes of data stored from location 3500h. [5]

OR

[B] Specify the register content and the flag status (S, Z, and CY) after the instruction ORA A is executed. [4]

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MVI A,A9H
MVI B, 57H
ADD B
ORA A
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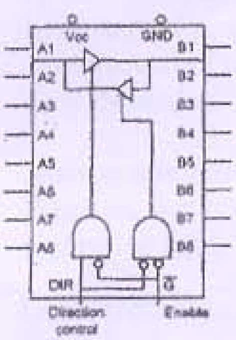
[C] Draw the circuit and Explain interface seven segment displays with microprocessor. [5]

Q-6 Answer the following questions.(Attempt any Two) **[12]**

[A] Draw and explain: Block diagram of 8254 Programmable interval timer.

[B] A string of six data bytes is stored starting from 2050H memory location. The string includes some Zero as mentioned below. Write a program along with flow chart to eliminate Zero from the string.
Data Value stored from 2050H location: C3,00,54,00,00,A2

[C] What is purpose of DMA? How is it interfaced with 8085? What are the internal components of DMA?



Function Table

Enable $\overline{G}$	Direction control DIR	Operation
L	L	
L	H	
H	X	

Figure 1